



Jack McGuire

Astro3510

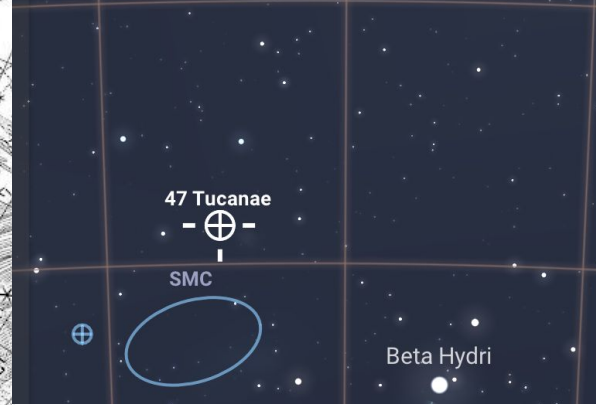
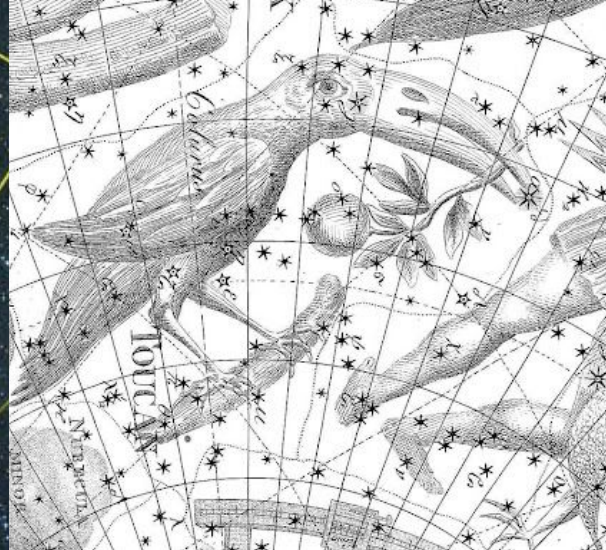
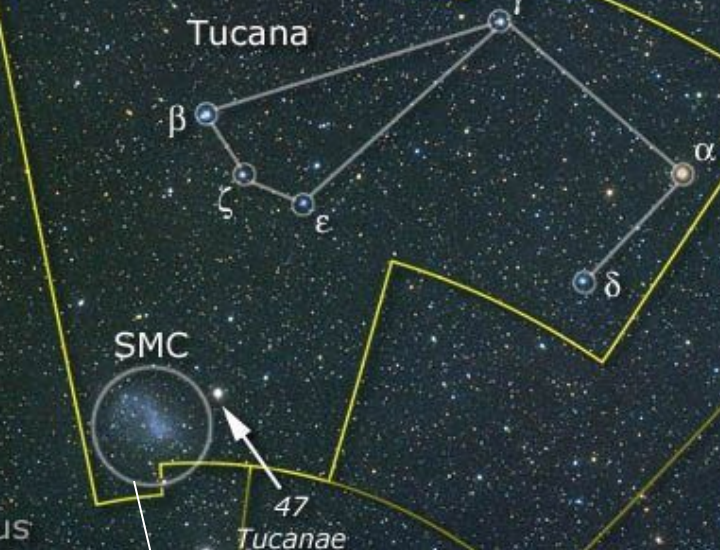
Presenting- October 24, 2024

Tuc-47, APOD Image- October 24, 2020

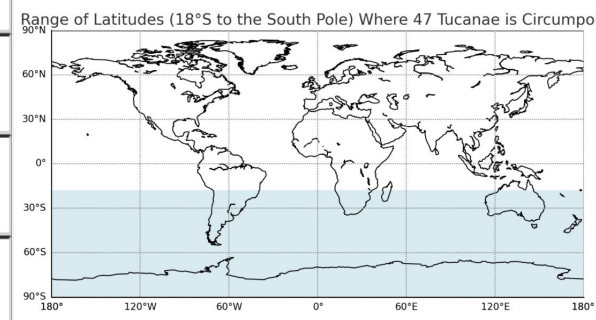
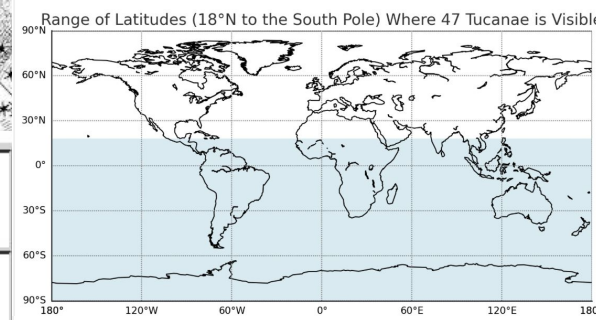
October 24, 2020 - Astronomy Picture Of The Day

Explanation: Globular star cluster 47 Tucanae is a jewel of the southern sky. Also known as NGC 104, it roams the halo of our Milky Way Galaxy along with some 200 other globular star clusters. The second brightest globular cluster (after Omega Centauri) as seen from planet Earth, it lies about 13,000 light-years away and can be spotted naked-eye close on the sky to the Small Magellanic Cloud in the constellation of the Toucan. The dense cluster is made up of hundreds of thousands of stars in a volume only about 120 light-years across. Red giant stars on the outskirts of the cluster are easy to pick out as yellowish stars in this sharp telescopic portrait. Tightly packed globular cluster 47 Tuc is also home to a star with the closest known orbit around a black hole.

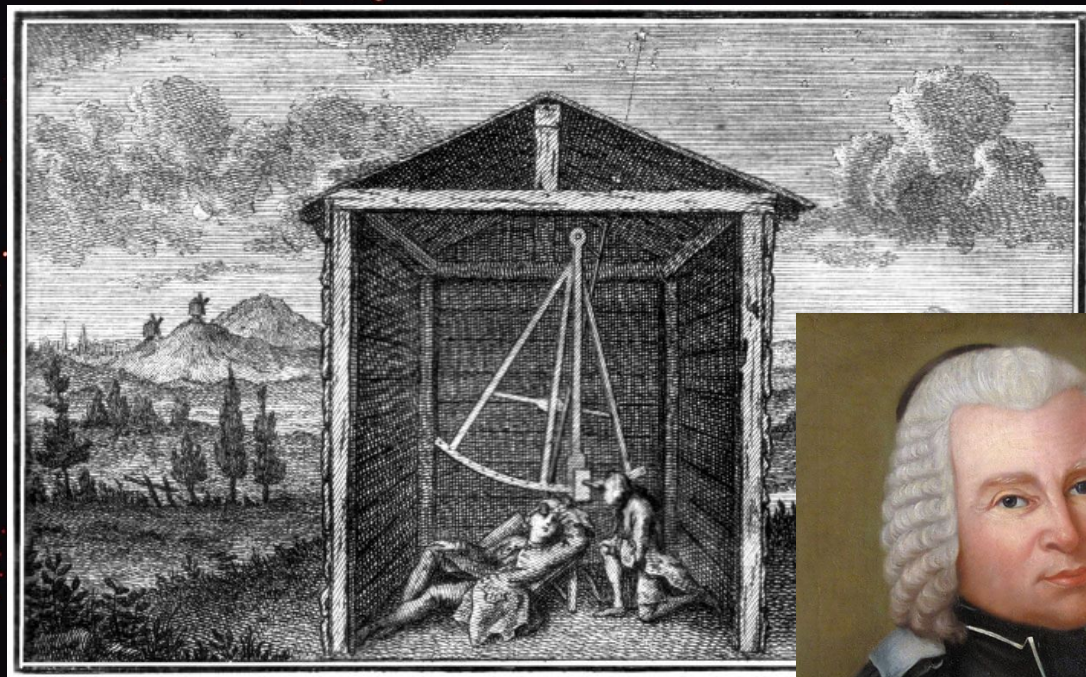




Right Ascension	00 : 24.1 (h:m)
Declination	-72 : 05 (deg:m)
Distance	13.4 (kly)
Visual Brightness	4.03 (mag)
Apparent Dimension	30.9 (arc min)



Nicholas Louis de Lacaille



Johann Bode



Globular Cluster vs. Open Cluster

Age and Star Population:

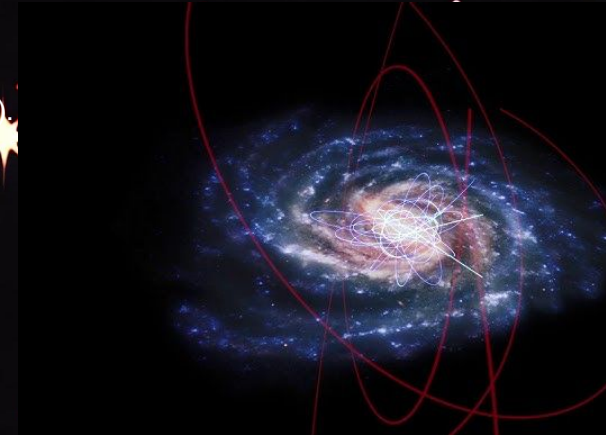
- Globular clusters: Old, metal-poor (Population II stars), 10-13 billion years old.
- Open clusters: Younger, metal-rich (Population I stars), tens to hundreds of millions of years old.

Size and Structure:

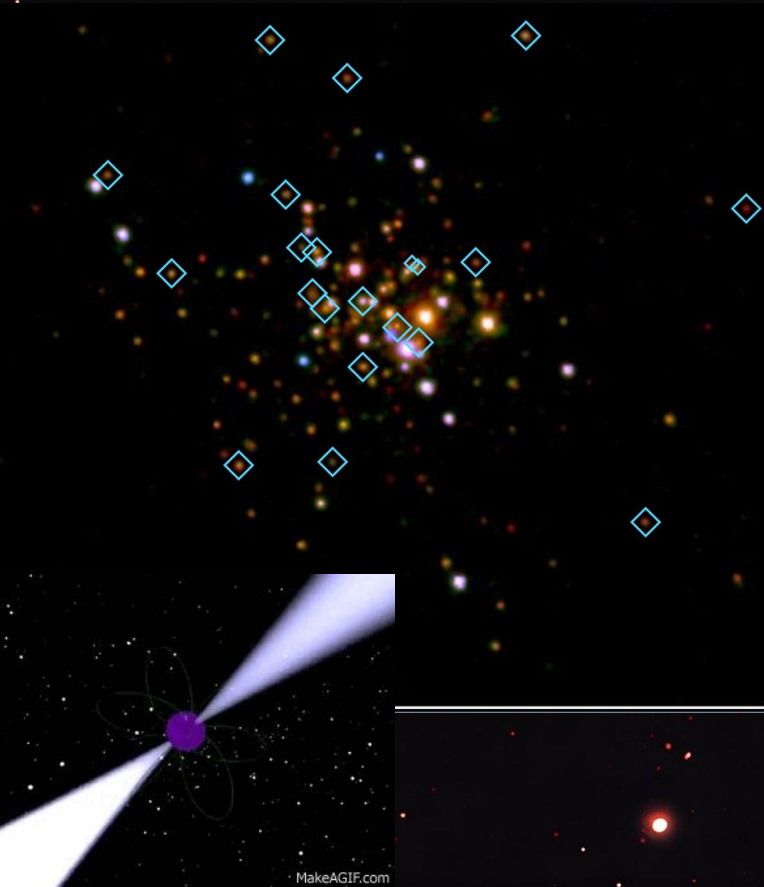
- Globular clusters: Dense, spherical, contain hundreds of thousands of stars.
- Open clusters: Loose, smaller, contain a few hundred to a few thousand stars.

Location:

- Globular clusters: Found in the galactic halo, far above or below the disk.
- Open clusters: Located in the galactic disk, typically in star-forming regions (spiral arms).



Composition



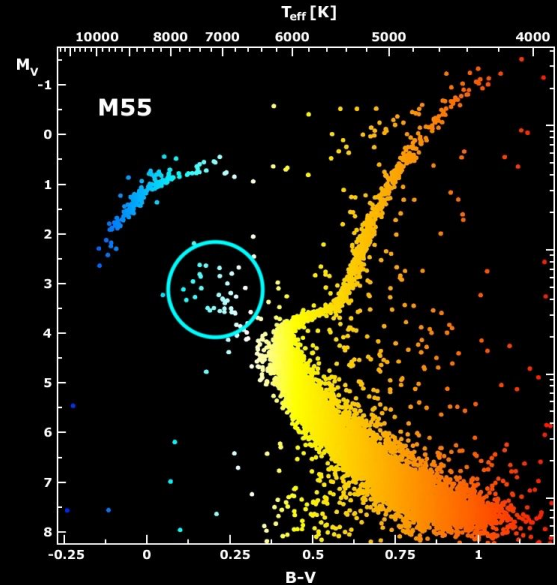
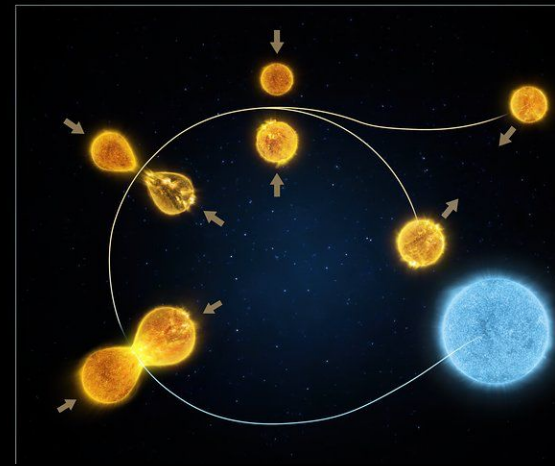
Main Sequence Stars: The most common stars in the cluster, burning hydrogen in their cores for billions of years.

Red Giants: Older, evolved stars that have expanded as they near the end of their life cycle, visible on the outskirts of the cluster.

White Dwarfs: The remnants of stars that have shed their outer layers, these faint, dense objects are scattered throughout the cluster.

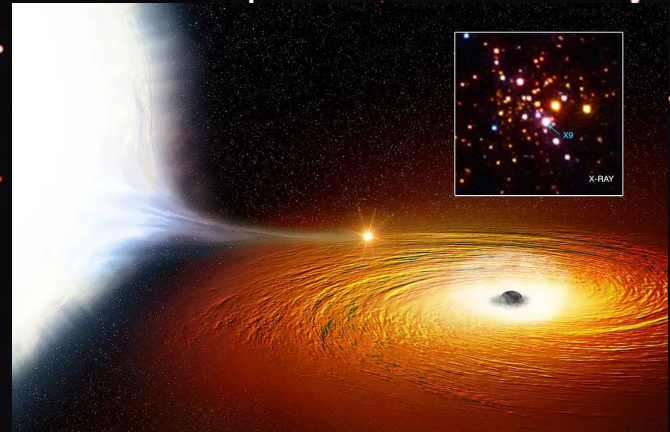
Blue Stragglers: Young, hot stars formed from the merger or collision of older stars, making them appear younger than the rest of the population.

Millisecond Pulsars: Rapidly spinning neutron stars, remnants of supernovae, known for their precise pulses of radiation.



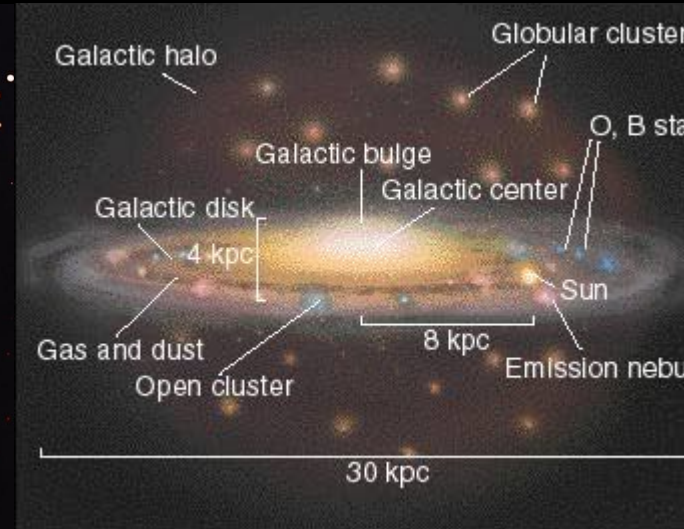
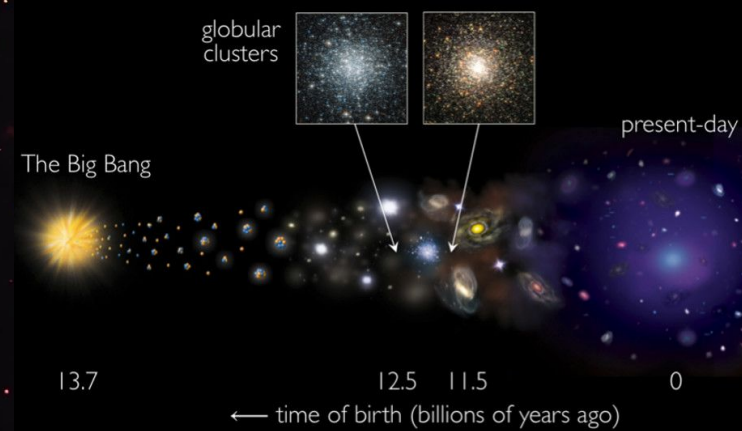
Closest Star to Orbit a Black Hole - X9

- **Closest known star orbiting a black hole** – just 28-minute orbital period.
- **White dwarf star** closely bound to a **stellar-mass black hole**.
- Detected via **X-ray emissions** from the system.
- **Orbital radius** is estimated to be about **1 million kilometers** (2.5 times the Earth-Moon distance).
- Black hole is **too small** to resolve directly using current techniques but is inferred
 - a. NOT the central black hole of Tuc 47



Formation

- **Ancient Relics:** Formed over **12 billion years ago**, a **fossil record** of the early universe.
- **Galactic Halo:** Shows that globular clusters formed **before** the **Milky Way's disk** flattened.
- **Merging History:** Its **elliptical orbit** suggests it may have been **captured from smaller galaxies**, hinting at the Milky Way's **merger history**.
- **Chemical Composition:** Low **metallicity** indicates formation from **primordial gas**, revealing early **star formation conditions**.



Mysteries left to explore with Globular Clusters

- **Orbital trajectory:** 47 Tucanae's orbit does not align with the galaxy's disk.
 - **Galactic mergers:** Could it have been captured from a smaller galaxy?
 - **Ejection from center:** Was it ejected from the Milky Way's core during formation?
- **Peculiar shape:** Why do globular clusters remain spherical while other systems flatten?
 - Do all globular clusters need one to maintain their structure?



Sources

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